A Sub-Pixel Coupled Dictionary Learning Method for Large-Scale and High-Spatial-Resolution Satellite Hyperspectral Image Reconstruction

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***Abstract*— Satellite hyperspectral images (HSIs) are often constrained by the limited spatial resolution and narrow imaging width, hindering precise classification in vast observation scenes despite boasting hundreds of approximately contiguous spectral channels and high spectral resolution. In contrast, satellite multispectral images (MSIs) lack spectral details but offer comprehensive spatial information in large scenes. To overcome the limitations of HSIs and MSIs, we proposed a novel MSI-HSI spatial-spectral collaborative representation model (Sub-CDL), leveraging sub-pixel representation and coupled dictionary learning. The model learns information consistency between MSI and HSI through the construction of a sub-pixel coupled dictionary, where spatial-resolution disparities between HSI and MSI are addressed by using the sub-pixel dictionary, spatial down-sampling matrix, and spatial blur matrix. Additionally, spectral-resolution differences are resolved by spectral response function (SRF). More specifically, the method utilizes spatial regularization constraint and sub-pixel coupled dictionary low-rank constraint as restrictive conditions, and the alternating direction method of multipliers (ADMM) is used to optimize the model. Ultimately, high-resolution reconstruction of HSIs with large scenes is enabled. By comparing with various state-of-the-art methods, experimental results demonstrate the proposed model's superiority in both authenticity and classification accuracy of reconstructed images.**

***Index Terms*— Dictionary learning, hyperspectral image (HSI), spatial super-resolution, sub-pixel representation.**

I. Introduction

COMPOSED of numerous approximately contiguous spectral channels, HS data provides extensive image and spectral details, boasting a high spectral resolution, and can achieve accurate identification and fine classification of various feature categories. At present, hyperspectral images (HSIs) are increasingly being utilized in various fields, including feature class identification [1], resource investigation [2], environmental monitoring [3], [4], [5], [6], meteorology [7], [8], [9], [10], agriculture [11], [12], [13], [14], [15], geology [16], military [17], [18], [19], [20], and numerous other domains with a wide range of promising application prospects. However, due to the constraints of the sensor's imaging ability and the law of energy conservation, satellite HSIs generally have low spatial resolution (LR) and narrow imaging swaths. In contrast, satellite multispectral images (MSIs) usually contain a few to dozens of bands, and they have higher spatial resolution compared with HSIs, but lower spectral resolution, resulting in limited ability to identify land covers. The limited spatial resolution of HSIs often leads to mixed pixels, which hinders precise data classification and accurate object recognition. Moreover, the narrow imaging width results in the limited image coverage of HSI, and large-scene HSI can only be acquired by stitching, which requires complicated preprocessing to minimize stitching errors and thus complicates data acquisition for large scenes. Consequently, to overcome the limitations of HSIs and MSIs, and capture spectral information from low-spatial-resolution HSIs (LR-HSIs) and spatial information from high-spatial-resolution MSIs (HR-MSIs), a combined utilization method of HSI and MSI is explored in the article to generate high-spatial-resolution HSIs (HR-HSIs) for large scenes.

At this stage, image fusion and image super-resolution are the two main implementation methods in the study of MS-HS co-utilization. The former generates fused HR-HSI with the same scene as the input LR-HSI [21], however, the latter is not limited by the input imaging width and can obtain large-scene HR-HSIs that are consistent with the scene of input HR-MSIs. Deep learning and dictionary learning are the primary super-resolution techniques. Deep learning methods learn the nonlinear MS-HS representation via a data-driven approach optimized by minimizing error and using back-propagation [22], [23], [24], [25]. However, the methods have limited generalization capabilities, rely on existing MSI-HSI and their performance is affected by image quality. Since the wavelength of multispectral sensors is roughly 400–1000 nm, while it is 400–2500 nm in hyperspectral sensors, thus the deep learning method cannot accurately reconstruct large-scene HR-HSI in the range of 1000–2500 nm. Dictionary learning extracts a spectral dictionary from LR-HSI, representing the reflectance. The dictionary comprises spectral vectors, which

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capture the properties of a specific object. By combining spectral dictionary with sparse representation coefficients from LR-HSI and HR-MSI, HR-HSI can be reconstructed in the wavelengths of visible and near-infrared [26].

Initially, Arad and Ben-Shahar [27] proposed a spectral super-resolution model based on sparse representation, using RGB and hyperspectral training data to estimate sparse coefficients and obtain HR-HSIs. Wu et al. [28] improved this method with a new shallow learning approach. Yokoya et al. [29] used sparse representation to recover large-scale HSIs from partially overlapping hyperspectral and multispectral data. Fotiadou et al. [30] proposed a coupled dictionary learning model that jointly learns low and high-spectral-resolution features to enhance spectral super-resolution. Gao et al. [31] adopted a joint sparse low-rank learning approach for more accurate reconstruction. Han et al. [32] introduced a spectral library to improve spectral super-resolution quality through joint utilization of multi-HSIs via dictionary learning. Liu et al. [33] proposed a class-guided coupled dictionary-learning method that incorporates training sample labels to create discriminative regularization terms for HSI reconstruction. Liu et al. [34] introduced a method that constructs class-specific and mutually coupled dictionaries using sample labels, emphasizing class-specificity for improved classification. To capitalize on HSI's inherent features such as spectral correlation, local sparsity, nonlocal self-similarity, and spatial piecewise smoothness, Rostami and Shirazi [35] decomposed the HSI-SR problem into multiple identical SC-SR tasks. Nie et al. [36] archived fine classification by maintaining data structures with low-rank and sparsity constraints introducing a discriminative incremental dictionary method to improve dictionary representation. The integration of coupled dictionaries and spectral libraries is also a current means of achieving spectral-SR. For example, to archive spectral-SR, Yan et al. [37] created a spectral library and fused MSI spatial information with its hyperspectral data to get a latent HSI. By learning a low-rank HSI dictionary, get an adaptive sparse MSI coefficients. Han et al. [38] utilized a typical set of spectra and the remaining spectral library as priors to establish a new spectral-SR model, and enhanced performance by optimizing the spectral dictionary. Leng et al. [39] proposed a spectral-SR method combining a universal and private spectral library for high-spatial MS to HS image conversion and the spectral dictionary and coefficients are optimized using alternating direction method of multipliers (ADMM).

Generally speaking, dictionary learning methods standardize reconstruction by manually setting priors to depict the HSI structure. However, both deep-learning and dictionary-learning methods overlook spatial resolution disparities between MSI and HSI, leading to information loss during preprocessing and compromising reconstruction quality.

In the article, we innovatively combine sub-pixel representation with the construction of coupled dictionaries and propose a novel large-scale and high-spatial-resolution satellite HSI reconstruction based on multi-hyperspectral collaborative representation. By considering spectral resolution disparities between HSI and MSI, we utilize a sub-pixel dictionary that combines HSI and MSI characteristics for learning, and the accuracy of the spatial reconstruction is enhanced by incorporating additional spatial information from overlapping MSI. Furthermore, since the narrower coverage of HSI compared to MSI, we design a sub-pixel coupled dictionary to effectively capture the coherence information between MSI-HSI, and finally a large-scale HR-HSI is reconstructed.

The main contributions of the article can be summarized as follows.

1) A sub-pixel coupled dictionary learning method Sub-CDL is proposed which combines the sub-pixel representation technique with the establishment of coupled dictionaries to address spatial and spectral resolution differences between HSI and MSI in their collaborative representation, thus realizing the accurate reconstruction of the large-scale HR-HSI.
2) Spectral-resolution differences are resolved by using spectral response function (SRF). To overcome the spatial-resolution differences between HSIs and MSIs, spatial blurred matric is innovatively incorporated with the down-sampling process, and together with the sparse representation coefficients derived from HR-MSIs, spatial correlations from high to low-resolution are established. Furthermore, by integrating sub-pixel dictionaries, a precise representation of LR-HSIs is achieved.
3) Spatial regularization constraints and sub-pixel coupled dictionary low-rank constraints are used to retain edges and detailed spatial information in images, thereby establishing a low-to-high spatial-spectral representation relationship for learning coherent information about MSI-HSI in overlapping regions. Subsequently, the ADMM is utilized to learn and optimize a sub-pixel dictionary.

The remainder of this article comprises four parts: Section II details the proposed method. Section III discusses datasets and evaluation metrics. Section IV analyzes the experimental results. Finally, Section V presents the conclusion of the work.

II. Proposed Method

In this section, we constructed a multi-hyperspectral collaborative sub-pixel representation model called Sub-CDL based on coupled dictionary learning. The consistency information between MSI and HSI is fully learned by designing coupled dictionaries, so as to complete an HR-HSI in large scenes covered by MSI.

A. Overall Structure of Sub-CDL

In Sub-CDL, to address the differences in spatial resolution and spectral resolution between LR-HSI and HR-MSI, we integrate spatial down-sampling and spatial blurred matrices to solve spatial resolution disparities in MS-HS data. We employ SRF to reconcile spectral resolution differences. Spatial regularization constraint and sub-pixel coupled dictionary low-rank constraint are utilized to facilitate the learning and optimization of multi-hyperspectral coupled dictionaries. As shown in Fig. 1, the proposed methodology proceeds as follows.

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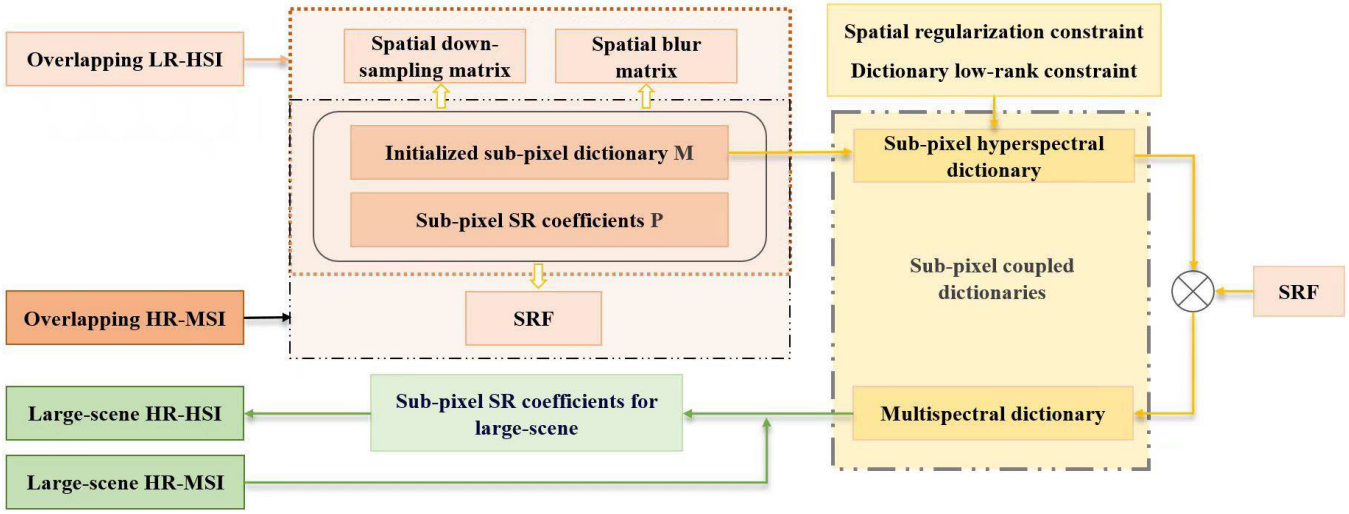


Fig. 1. Overall structure of Sub-CDL.

1) Initially, overlapping MSI and HSI are leveraged to establish a coupled sub-pixel dictionary learning model, to fully utilize the spectral and spatial information in LR-HSI and HR-MSI and learn a sub-pixel HS dictionary. The sub-pixel HS dictionary is firstly initialized, and spatial down-sampling matrix and spatial blur matrix are used to connect the information in LR-HSI with the sub-pixel HS dictionary, SRF is used to connect the sub-pixel HS dictionary with the information in HR-MSI. With spatial regularization constraint and sub-pixel dictionary low-rank constraint, the model is subsequently optimized through the ADMM. The approach enables the comprehensive learning of the coupled sub-pixel dictionary and the extraction of collaborative features from MS-HS data.
2) Subsequently, SRF is employed to craft a multispectral dictionary. The sub-pixel representation coefficient within extensive scenes is learned by integrating large-scene MSI. The process enables high-resolution reconstruction of HSI within large scenes, thus refining and enhancing the quality of HSIs.

B. Coupled Sub-Pixel Dictionary Learning Model

Firstly, reorganize LR-HSI in overlapping regions into a 2-D matrix consisting of single-pixel vectors $\mathbf{Y}_h = [y_{h1}, \ldots, y_{hn}] \in \mathbb{R}^{b_h \times n}$, where b_h represents the number of bands of LR-HSI and n represents the number of pixels in LR-HSI. We defined $\mathbf{M} = [m_1, \ldots, m_p] \in \mathbb{R}^{b_h \times p}$ as an initialized sub-pixel hyperspectral dictionary to be learned and updated which has p dictionary atoms and b_h spectral channels.

Furthermore, $\mathbf{S} = [s_1, \ldots, s_n] \in \mathbb{R}^{p \times n}$ is defined to represent the sparse representation coefficient for HS data $\mathbf{Y}_h$ with n pixels, and $\mathbf{S}$ needs to satisfy the constraints: $\mathbf{S} \geq 0$ and $\mathbf{1}_p^{\mathrm{T}}\mathbf{S} = \mathbf{1}_n^{\mathrm{T}}$, $\mathbf{1}_p^{\mathrm{T}} = [1, 1, \ldots, 1]^{\mathrm{T}}$, which keeps the sparse representation coefficient matrix non-negative and ensures the sum of all coefficients is 1, respectively. Adding spatial resolution differences between HSI and MSI into consideration, $\mathbf{P} = [p_1, \ldots, p_v] \in \mathbb{R}^{p \times v}$ can be introduced as the sub-pixel sparse representation coefficient matrix for HS data, where the number of sub-pixels in $\mathbf{P}$ is $v = n \times s^2$, and s is a scaling factor. Assuming $\mathbf{D} = [d_1, \ldots, d_n] \in \mathbb{R}^{v \times n}$ as a spatial down-sampling matrix, we can establish a precise sub-pixel representation relationship $\mathbf{S} = \mathbf{PD}$ between the sparse and the sub-pixel sparse representation coefficient.

Subsequently, by incorporating the sub-pixel representation relationship along with the spatial blur matrix and the spatial down-sampling matrix, and taking the fidelity term of HS data into account, we can derive the following sub-pixel representation model in overlapping LR-HSI:

$$\mathbf{Y}_h = \mathbf{MPBD} + \mathbf{N}_1 \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{v \times v}$ is a blurring matrix on a spatial dimension, namely a point expansion function during the real imaging process, which can be estimated from the overlapping MSI-HSI [40]; $\mathbf{P} = [p_1, \ldots, p_v] \in \mathbb{R}^{p \times v}$ represents the sparse coefficient for sub-pixels. $\mathbf{N}_1$ means the additive noise.

If the average filtering method is adopted, then $\mathbf{D}$ can be defined as

$$\mathbf{D} = \frac{\left(\mathbf{I}_l \otimes \mathbf{1}_s^{\mathrm{T}}\right)^{\mathrm{T}} \otimes \left(\mathbf{I}_r \otimes \mathbf{1}_s^{\mathrm{T}}\right)^{\mathrm{T}}}{s^2} \tag{2}$$

where l and r represent the sizes of rows and columns in the original HSI, respectively. The number of pixels n in HSI satisfies $n = l \times r$, $\otimes$ represents the Cronecker product, and $\mathbf{I}$ represents the identity matrix of the corresponding size.

The overlapping MSI and HSI have simultaneous differences in spatial and spectral resolution. For HR-MSI of the overlapping areas, $\mathbf{Y}_m = [y_{m1}, \ldots, y_{mn}] \in \mathbb{R}^{b_m \times v}$ represents an image with b_m bands and v pixels, where $b_m < b_h$, $v = n \times s^2$. s is a scaling factor. $\mathbf{M}'$ represents a multispectral dictionary. The overlapping MSI can also be represented as

$$\mathbf{Y}_m = \mathbf{M}'\mathbf{P} + \mathbf{N}_2 \tag{3}$$

where $\mathbf{M}' = \mathbf{RM}$ represents the relationship between the HS and MS dictionary. $\mathbf{R} \in \mathbb{R}^{b_m \times b_h}$ is a spectral down-sampling matrix, which can be calculated from the SRF of

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the HS sensor. Similarly, $\mathbf{N}_2$ represents the additive noise in the process.

Due to the ill-posedness of the problem, spatial prior information is introduced to regularize the model in the task of image reconstruction

$$\varphi\left(\nabla_x\mathbf{P}, \nabla_y\mathbf{P}\right) \stackrel{\text{def}}{=} \sum_{j=1}^{v}\sqrt{\sum_{i=1}^{p}\left\{\left[\left(\nabla_x\mathbf{P}\right)_{ij}\right]^2 + \left[\left(\nabla_y\mathbf{P}\right)_{ij}\right]^2\right\}} \tag{4}$$

where $(\nabla_x\mathbf{P})_{ij}$ represents the elements in the row i and column j of the matrix, $\nabla_x\mathbf{P}$ and $\nabla_y\mathbf{P}$ represent the image horizontal and vertical discretization difference calculated by using the periodic boundary conditions, respectively. The spatial prior information retains the edges and the detailed spatial information in the image, aiming at imposing sparsity in the distribution of the absolute gradient of the image, which means that the transition between the pixels of the image should be smooth in the spatial dimension, except for a few pixels consistent with the edges.

Based on the considerations mentioned above, the proposed method of coupled sub-pixel dictionary learning can be represented by the following optimization problems:

$$\min_{\mathbf{P},\mathbf{M}}\left\{\begin{array}{l}\dfrac{1}{2}\left\|\mathbf{Y}_h - \mathbf{MPBD}\right\|_F^2 + \dfrac{\lambda}{2}\left\|\mathbf{Y}_m - \mathbf{RMP}\right\|_F^2 \\ \quad +\mu\varphi\left(\nabla_x\mathbf{P}, \nabla_y\mathbf{P}\right) + \gamma\left\|\mathbf{M}\right\|_*\end{array}\right\} \tag{5}$$

where the first two terms are data fidelity, the latter two are the spatial regularization constraint on representation coefficient and the low rank constraint on coupled subpixel dictionary, respectively; $\|\|_F^2$ and $\|\|_*$ represent the calculation of the F norm and kernel norm of the matrix, respectively.

C. Model Optimization Based on ADMM

In the proposed method, we optimize the problem through variable separation by using ADMM. Firstly, introduce auxiliary variables to divide the original optimization variable $\mathbf{P}$ into a total of six variables. One is still called $\mathbf{P}$, five auxiliary variables $\mathbf{U}_1$, $\mathbf{U}_2$, $\mathbf{U}_3$, $\mathbf{U}_4$, $\mathbf{J}$, respectively. The optimization problem is as follows:

$$\begin{aligned}\min_{\mathbf{P},\mathbf{M},\mathbf{U}_1,\mathbf{U}_2,\mathbf{U}_3,\mathbf{U}_4,\mathbf{J}}&\left\{\begin{array}{l}\dfrac{1}{2}\left\|\mathbf{Y}_h - \mathbf{MU}_1\mathbf{D}\right\|_F^2 \\ \quad +\dfrac{\lambda}{2}\left\|\mathbf{Y}_m - \mathbf{RMU}_2\right\|_F^2 \\ \quad +\mu\varphi\left(\mathbf{U}_3, \mathbf{U}_4\right) + \gamma\left\|\mathbf{J}\right\|_*\end{array}\right\} \\ \text{s.t. }&\mathbf{U}_1 = \mathbf{PB}, \quad \mathbf{U}_2 = \mathbf{P}, \quad \mathbf{U}_3 = \nabla_x\mathbf{P} \\ &\mathbf{U}_4 = \nabla_y\mathbf{P}, \quad \mathbf{J} = \mathbf{M}.\end{aligned} \tag{6}$$

The augmented Lagrangian function of the optimization model is

$$\begin{aligned}&\mathcal{L}\left(\mathbf{P}, \mathbf{M}, \mathbf{U}_1,\ \mathbf{U}_2, \mathbf{U}_3, \mathbf{U}_4, \mathbf{J}, \mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3, \mathbf{A}_4, \Lambda\right) \\ &= \frac{1}{2}\left\|\mathbf{Y}_h - \mathbf{MU}_1\mathbf{D}\right\|_F^2 + \frac{\mu}{2}\left\|\mathbf{PB} - \mathbf{U}_1 - \mathbf{A}_1\right\|_F^2 \\ &\quad + \frac{\lambda_m}{2}\left\|\mathbf{Y}_m - \mathbf{RMU}_2\right\|_F^2 + \frac{\mu}{2}\left\|\mathbf{P} - \mathbf{U}_2 - \mathbf{A}_2\right\|_F^2 \\ &\quad + \lambda_\varphi\varphi\left(\mathbf{U}_3, \mathbf{U}_4\right) + \frac{\mu}{2}\left\|\nabla_x\mathbf{P} - \mathbf{U}_3 - \mathbf{A}_3\right\|_F^2 \\ &\quad + \frac{\mu}{2}\left\|\nabla_y\mathbf{P} - \mathbf{U}_4 - \mathbf{A}_4\right\|_F^2 \\ &\quad + \gamma\left\|\mathbf{J}\right\|_* + \frac{\sigma}{2}\left\|\mathbf{J} - \mathbf{M}\right\|_F^2 + \Lambda^{\mathrm{T}}\left(\mathbf{J} - \mathbf{M}\right) + l_R^+\left(\mathbf{J}\right)\end{aligned} \tag{7}$$

where $\mathbf{A}_1 \sim \mathbf{A}_4$ and Λ denotes the Lagrangian multiplier, $\|\|_*$ represents the kernel norm of the matrix. $l_R^+()$ indicates the positive operator of elements, realizing the truncation of positive elements in the matrix

$$\begin{aligned}\mathbf{P}^{(k+1)} = \arg\min_{\mathbf{P}}&\frac{\mu}{2}\left\|\mathbf{PB} - \mathbf{U}_1^{(k)} - \mathbf{A}_1^{(k)}\right\|_F^2 \\ &+ \frac{\mu}{2}\left\|\mathbf{P} - \mathbf{U}_2^{(k)} - \mathbf{A}_2^{(k)}\right\|_F^2 \nabla_x\mathbf{P} \\ &+ \frac{\mu}{2}\left\|\nabla_x\mathbf{P} - \mathbf{U}_3^{(k)} - \mathbf{A}_2^{(k)}\right\|_F^2 \\ &+ \frac{\mu}{2}\left\|\nabla_y\mathbf{P} - \mathbf{U}_4^{(k)} - \mathbf{A}_3^{(k)}\right\|_F^2.\end{aligned} \tag{8}$$

Let the derivative of the function zero, we can get the solution

$$\begin{aligned}\mathbf{P}^{(k+1)} = &\left[\mathbf{BB}^{\mathrm{T}} + \mathbf{I} + \nabla_x\nabla_x^{\mathrm{T}} + \nabla_y\nabla_y^{\mathrm{T}}\right]^{-1} \\ &\cdot\left[\left(\mathbf{U}_1^{(k)} + \mathbf{A}_1^{(k)}\right)\mathbf{B}^{\mathrm{T}} + \left(\mathbf{U}_2^{(k)} + \mathbf{A}_2^{(k)}\right)\right. \\ &\left.+\left(\mathbf{U}_3^{(k)} + \mathbf{A}_3^{(k)}\right)\nabla_x^{\mathrm{T}} + \left(\mathbf{U}_4^{(k)} + \mathbf{A}_4^{(k)}\right)\nabla_y^{\mathrm{T}}\right].\end{aligned} \tag{9}$$

Within (9), the computation of image gradients necessitates the manipulation of extensive matrices. Considering the operational memory, processing velocity, and the periodically blocked cyclic characteristic of the gradient calculation matrix. The employment of the fast Fourier transform (FFT) serves as an effective tool in addressing the calculation, thereby mitigating the computational complexity to $O(p \times v\log(v))$.

Get the optimization function about the sub-pixel dictionary by separating variable M

$$\begin{aligned}\mathbf{M}^{(k+1)} = \arg\min_{\mathbf{M}}&\frac{1}{2}\left\|\mathbf{Y}_h - \mathbf{MU}_1^{(k)}\mathbf{D}\right\|_F^2 \\ &+ \frac{\lambda_m}{2}\left\|\mathbf{Y}_m - \mathbf{RMU}_2^{(k)}\right\|_F^2 \\ &+ \frac{\sigma}{2}\left\|\mathbf{J}^{(k)} - \mathbf{M}\right\|_F^2 + \Lambda^{\mathrm{T}}\left(\mathbf{J}^{(k)} - \mathbf{M}\right).\end{aligned} \tag{10}$$

The minimization function of $\mathbf{U}_1$ is a quadratic problem

$$\begin{aligned}\mathbf{U}_1^{(k+1)} = \arg\min_{\mathbf{U}_1}&\frac{1}{2}\left\|\mathbf{Y}_h - \mathbf{MU}_1\mathbf{D}\right\|_F^2 \\ &+ \frac{\mu}{2}\left\|\mathbf{P}^{(k+1)}\mathbf{B} - \mathbf{U}_1 - \mathbf{A}_1^{(k)}\right\|_F^2.\end{aligned} \tag{11}$$

For the down-sampling operation in the process, a mask matrix can be adopted to avoid the large memory occupied by the down-sampling matrix. A mask matrix $\mathbf{D}$ is used to decompose $\mathbf{U}_1$ into $\mathbf{U}_1\mathbf{D}$ and $\mathbf{U}_1\bar{\mathbf{D}}$, where $\bar{\mathbf{D}}$ indicates the pixel location not selected by $\mathbf{D}$ in the image

$$\begin{aligned}\mathbf{U}_1^{(k+1)}\mathbf{D} = &\left[\mathbf{M}^{\mathrm{T}}\mathbf{M} + \mu\mathbf{I}\right]^{-1} \\ &\times\left[\mathbf{M}^{\mathrm{T}}\mathbf{Y}_h + \mu\left(\mathbf{P}^{(k+1)}\mathbf{B} - \mathbf{A}_1^{(k)}\right)\right]\mathbf{D}\end{aligned} \tag{12}$$

$$\mathbf{U}_1^{(k+1)}\bar{\mathbf{D}} = \left(\mathbf{P}^{(k+1)}\mathbf{B} - \mathbf{A}_1^{(k)}\right)\bar{\mathbf{D}}. \tag{13}$$

Moreover, the optimization of variable $\mathbf{U}_2$ is also a minimized quadratic problem. By taking the derivative of the

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objective function and keeping the derivative zero, the solution of the variable can be shown in (14). The computational complexity is $O(p \times v)$

$$\begin{aligned}\mathbf{U}_2^{(k+1)} &= \arg\min_{\mathbf{U}_2} \frac{\lambda_m}{2} \|\mathbf{Y}_m - \mathbf{RMU}_2\|_F^2 \\ &\quad + \frac{\mu}{2} \left\|\mathbf{P}^{(k+1)} - \mathbf{U}_2 - \mathbf{A}_2^{(k)}\right\|_F^2 \\ &= \left[\lambda_m \mathbf{M}^{\mathrm{T}}\mathbf{R}^{\mathrm{T}}\mathbf{RM} + \mu\mathbf{I}\right]^{-1} \\ &\quad \times \left[\lambda_m \mathbf{M}^{\mathrm{T}}\mathbf{R}^{\mathrm{T}}\mathbf{Y}_m + \mu\left(\mathbf{P}^{(k+1)} - \mathbf{A}_2^{(k)}\right)\right]. \end{aligned} \tag{14}$$

Equation (15) represents the optimization function of $\mathbf{U}_3$, $\mathbf{U}_4$. Since the formula contains the constraint terms representing the spatial prior information, the variables can be jointly solved using the pixel-by-pixel soft threshold function. The computational complexity is $O(p \times v \log(v))$

$$\begin{aligned}\mathbf{U}_3^{(k+1)}, \mathbf{U}_4^{(k+1)} &= \arg\min_{\mathbf{U}_3, \mathbf{U}_4} \lambda_\varphi \varphi\left(\mathbf{U}_3, \mathbf{U}_4\right) \\ &\quad + \frac{\mu}{2} \left\|\nabla_x \mathbf{P}^{(k+1)} - \mathbf{U}_3 - \mathbf{A}_3^{(k)}\right\|_F^2 \\ &\quad + \frac{\mu}{2} \left\|\nabla_y \mathbf{P}^{(k+1)} - \mathbf{U}_4 - \mathbf{A}_4^{(k)}\right\|_F^2. \end{aligned} \tag{15}$$

The introduction of variable $\mathbf{J}$ ensures a low-rank constraint on the sub-pixel dictionary. Similarly, $\|\|_*$ represents the kernel norm of a matrix, and $l_R^+()$ represents the positive operator of the matrix elements, which realizes the truncation of the positive elements in the matrix. What's more, the optimization of $\mathbf{J}$ can be effectively solved by using singular value thresholding (SVT)

$$\begin{aligned}\mathbf{J}^{(k+1)} &= \arg\min_{\mathbf{J}} \gamma \|\mathbf{J}\|_* + \frac{\sigma}{2} \|\mathbf{J} - \mathbf{M}\|_F^2 \\ &\quad + \Lambda^{\mathrm{T}}\left(\mathbf{J} - \mathbf{M}\right) + l_R^+\left(\mathbf{J}\right). \end{aligned} \tag{16}$$

In the singular value threshold operator, firstly, singular value decomposition is performed on the input low-rank matrix, assuming the rank of the input matrix $\mathbf{G}$ is r, there is

$$\mathrm{SVD}\left(\mathbf{G}\right) \approx \mathbf{USV}, \quad \mathbf{S} = \mathrm{diag}\left(\{s_k\}_{1 \le k \le r}\right). \tag{17}$$

For the arbitrary $\tau \ge 0$, introducing the soft threshold operator D_r

$$D\left(\mathbf{G}\right) = \mathbf{U}D_r\left(\mathbf{S}\right)\mathbf{V}, \quad D_r\left(\mathbf{S}\right) = \mathrm{diag}\left(\{s_k - \tau\}^+\right). \tag{18}$$

Then, $\|\mathbf{G}\|_*$ can be expressed as $\|D_r(\mathbf{S})\|_{1,1}$, hence the update mode of the auxiliary variable $\mathbf{J}$ can be expressed as

$$\mathbf{J}^{k+1} = \max\left\{\mathbf{0}, D_r\left(\mathbf{M} - \mathbf{\Lambda}/\sigma\right)\right\}. \tag{19}$$

The Lagrange multipliers are updated as follows:

$$\mathbf{A}_1^{(k+1)} = \mathbf{A}_1^{(k)} - \left(\mathbf{P}^{(k+1)}\mathbf{B} - \mathbf{U}_1^{(k+1)}\right) \tag{20}$$

$$\mathbf{A}_2^{(k+1)} = \mathbf{A}_2^{(k)} - \left(\mathbf{P}^{(k+1)} - \mathbf{U}_2^{(k+1)}\right) \tag{21}$$

$$\mathbf{A}_3^{(k+1)} = \mathbf{A}_3^{(k)} - \left(\nabla_x \mathbf{P}^{(k+1)} - \mathbf{U}_3^{(k+1)}\right) \tag{22}$$

$$\mathbf{A}_4^{(k+1)} = \mathbf{A}_4^{(k)} - \left(\nabla_y \mathbf{P}^{(k+1)} - \mathbf{U}_4^{(k+1)}\right) \tag{23}$$

$$\Lambda^{(k+1)} = \Lambda + \sigma\left(\mathbf{J}^{(k+1)} - \mathbf{M}\right). \tag{24}$$

Algorithm 1 ADMM

Input: LR-HSI $\mathbf{Y}_h$, HR-MSI $\mathbf{Y}_m$, Spectral response function $\mathbf{R}$, Spatial blur matrix $\mathbf{B}$, Mask matrix $\mathbf{D}$;
Initialization: $\lambda_m = 1$, $\lambda_\varphi = 10^{-3}$, $\sigma = 0.1$, $\mu = 0.05$, $\gamma = 1$, $\varepsilon = 10^{-6}$, $k = 1$, max Iter $= 1000$;
$\mathbf{U_1} \sim \mathbf{U_4}$, $\mathbf{J}$, $\mathbf{A}_1 \sim \mathbf{A}_4$, Λ, $\mathbf{M}$, $\mathbf{P}$ are initialized as zero matrix;
While $k <$ max Iter do:
  Update $\mathbf{P}$ via (9);
  Update $\mathbf{M}$ via (10);
  Update $\mathbf{U_1} \sim \mathbf{U}_4$ via (11)-(15);
  Update $\mathbf{J}$ via (16);
  Update $\mathbf{A}_1 \sim \mathbf{A}_4$, Λ via (20)-(24);
  if $\|\mathbf{P}^{t+1} - \mathbf{P}^t\|_F^2 < \varepsilon$ do:
    Stop iterating
  else do:
    $k \leftarrow k + 1$
Output: Sub-pixel dictionary $\mathbf{M}$

D. Reconstruction of HR-HSI in a Large Scene

For the optimized sub-pixel dictionary, the large scene HR-MSI can be sparsely represented by the model in the following formula:

$$\min_{\mathbf{X}} \frac{1}{2} \|\mathbf{Y}_{\mathrm{in}} - \mathbf{RMX}\|_F^2 + \eta \|\mathbf{X}\|_{1,1}, \quad \text{s.t. } \mathbf{1}^{\mathrm{T}}\mathbf{X} = 1 \tag{25}$$

where $\mathbf{Y}_{\mathrm{in}}$ represents the input large-scene MSI, $\mathbf{X}$ indicates the sparse representation coefficients encoded on MSI by using the sub-pixel dictionary. $\|\mathbf{X}\|_{1,1}$ represents a sparse constraint on representation coefficients. The sum of sparse representation coefficients is constrained to be 1 in $\mathbf{1}^{\mathrm{T}}\mathbf{X} = 1$. $\mathbf{R} \in \mathbb{R}^{b_m \times b_h}$ is a spectral down-sampling matrix, which can be calculated from the SRF of the HSI sensor.

Based on the optimization model, auxiliary variables are introduced to obtain the following augmented Lagrangian function:

$$\begin{aligned}\mathcal{L}\left(\mathbf{X}, \mathbf{O}, \mathbf{\Delta}\right) &= \frac{1}{2} \|\mathbf{Y}_{\mathrm{in}} - \mathbf{RMX}\|_F^2 + \eta \|\mathbf{O}\|_{1,1} \\ &\quad + \mathbf{\Delta}^{\mathrm{T}}\left(\mathbf{O} - \mathbf{X}\right) + \frac{\rho}{2} \|\mathbf{O} - \mathbf{X}\|_F^2 \\ &\quad \text{s.t. } \mathbf{1}^{\mathrm{T}}\mathbf{X} = \mathbf{1} \end{aligned} \tag{26}$$

where $\mathbf{\Delta}$ and ρ represent the Lagrange multiplier and Lagrangian penalty parameters, respectively.

After applying ADMM and fixing other variables, the sparse coefficient $\mathbf{X}$ can be optimized by solving the constrained ordinary least squares regression

$$\begin{aligned}&\min_{\mathbf{X}} \frac{1}{2} \|\mathbf{Y}_{\mathrm{in}} - \mathbf{RMX}\|_F^2 + \mathbf{\Delta}^{\mathrm{T}}\left(\mathbf{O} - \mathbf{X}\right) + \frac{\rho}{2} \|\mathbf{O} - \mathbf{X}\|_F^2 \\ &\text{s.t. } \mathbf{1}^{\mathrm{T}}\mathbf{X} = \mathbf{1}. \end{aligned} \tag{27}$$

Solving (27) yields an analytical solution

$$\mathbf{X} \leftarrow \mathbf{A}^{-1}\mathbf{B} - \mathbf{C}\left(\mathbf{I}_{1 \times L}\mathbf{A}^{-1}\mathbf{B} - 1\right) \tag{28}$$

where $\mathbf{I}_{1 \times L}$ represents a column vector of all 1s with a length of L. The remaining intermediate variables are expressed as

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follows:

$$\mathbf{A} = \left(\mathbf{R}\mathbf{M}^{\mathrm{T}}\right)\mathbf{R}\mathbf{M} + \rho\mathbf{I}_{L\times L} \tag{29}$$

$$\mathbf{B} = \mathbf{R}\mathbf{M}^{\mathrm{T}}\mathbf{Y}_{\mathrm{in}} + \rho\mathbf{O} + \boldsymbol{\Delta} \tag{30}$$

$$\mathbf{C} = \mathbf{A}^{-1}\mathbf{I}_{L\times 1}\left(\mathbf{I}_{1\times L}\mathbf{A}^{-1}\mathbf{I}_{L\times 1}\right)^{-1}. \tag{31}$$

The solution for the auxiliary variable $\mathbf{O}$ can be obtained through variable separation, leading to the following minimization problem:

$$\min_{\mathbf{O}} \eta \left\|\mathbf{O}\right\|_{1,1} + \boldsymbol{\Delta}^{\mathrm{T}}\left(\mathbf{O}-\mathbf{X}\right) + \frac{\rho}{2}\left\|\mathbf{O}-\mathbf{X}\right\|_F^2. \tag{32}$$

The l_1 norm of variable $\mathbf{O}$ is involved in the optimization problem, which can be solved by the soft threshold operator

$$\mathbf{O} \leftarrow \operatorname{sign}\left(\mathbf{X} - \frac{\boldsymbol{\Delta}}{\rho}\right) \odot \max\left\{\mathbf{0}, \left|\mathbf{X} - \frac{\Delta}{\rho}\right| - \frac{\eta}{\rho}\right\} \tag{33}$$

where $\odot$ represents a Hadamard product of the matrix and sign() is a symbolic function.

The Lagrange multiplier is updated as follows:

$$\boldsymbol{\Delta} \leftarrow \boldsymbol{\Delta} + \rho\left(\mathbf{O}-\mathbf{X}\right). \tag{34}$$

Under the assumption that the overlapped MSI-HSI share the same set of representation coefficients, the reconstructed HR-HSI can be generated by (35) via sparse representation coefficients obtained based on MSI

$$\mathbf{Y}_{\mathrm{out}} = \mathbf{M}\mathbf{X}. \tag{35}$$

III. Experimental Datasets and Evaluation Indicators

A. Experimental Datasets

We use Chikusei data and GF-YR data in the experiment. The Chikusei data contains HSIs of Chikusai, Ibaraki, Japan, acquired by the Hyperspec-VNIR-CIRIS spectrometer. With a ground sampling distance of 2.5 m, the image size is 2517 × 2335 pixels. Chikusei contains 128 bands in the spectral range from 363 to 1018 nm. The dataset collected ground truth for 16 categories in the selected experimental range.

In the experiment, we cropped the Chikusei HSI to a size of 1290 × 1508, and 16 categories are contained in the cropped image. As shown in Fig. 2(a), the original HSI is displayed as bands 64, 42, and 23 for channels R, G, and B, respectively. And we got the HR-MSI and LR-HSI by spectrally and spatially down-sampling the original HSI. Specifically, considering the impacts of factors such as atmospheric conditions and sensor noise during the imaging process, blurring operation inspired by Hysure [40] was employed. Utilizing the Starck-Murtagh filter [41] for spatial blur, in order to simulate various types of interference present in actual imaging processes. Subsequently, the blurred HR-HSI was spatial down-sampled to get a LR-HSI. Assuming that the full width at half maximum (FWHM) is 6, the SRF is simulated by the Gaussian function centered on wavelength. Subsequently, the indicators of MS and HS Chikusei data, such as bands, source, and image size, are listed in Table I. Moreover, Fig. 2(d) and (e) show the training set and test set labels. The number of labels for training sets and test sets during the classification of Chikusei data is listed in Table III.

TABLE I
INDICATORS OF CHIKUSEI MS-HS DATA

Dataset	Source	Image size	Number of bands	Spatial resolution
Original HR-HSI	Hyperspec-VNIR-CIRIS	1290×1508	128	2.5m
HR-MSI	Simulate	1290×1508	10	2.5m
LR-HSI	Simulate	645×754	128	5m

TABLE II
INDICATORS OF GF-YR MS-HS DATA

Dataset	Source	Image size	Number of bands	Spatial resolution
HR-MSI	GF-1	1468×1526	4	15m
LR-HSI	GF-5	734×763	295	30m

TABLE III
NUMBER OF LABELS FOR TRAINING AND TEST SETS

Labels	Categories	Number of Training Labels	Number of Testing Labels
1	Rice field (first stage)	76	25332
2	Natural plant	1312	68922
3	Row crops	302	77755
4	Asphalt	809	113088
5	Forest	402	151984
6	Water	739	25161
7	Plastic House	432	11112
8	Bare soil (school)	211	18928
9	Bare soil (farmland)	273	225161
10	Manmade grass	450	47367
11	Paved ground	223	56771
12	Manmade (dark)	312	60301
13	Weeds in farm	217	102359
14	Bare soil (park)	422	32929
15	grass	422	244597
16	Rice field (grown)	380	650121

The HS and MS data of GF-YR are acquired by GF-5 and GF-1 satellites, respectively. As shown in Table II, a 16 m multispectral camera carried by GF-1 can provide MS data with four bands at the ranges of 450–520 nm, 520–590 nm, 630–690 nm, and 770–890 nm.

The GF-5 satellite's full-band spectral imager captures imaging data with a spatial resolution of 30 m. The imaging spectrum spans a broad range from 400 to 2500 nm, encompassing a total of 330 bands. Notably, spectral resolution of the visible band can achieve 5 nm. After preprocessing the data to eliminate defective bands, the image retains 295 bands for analysis. Fig. 3 shows the false-color display of GF-5 LR-HSI in bands 56, 39, and 25 and GF-1 HR-MSI in bands 3, 2, and 1, respectively. The number of labels for each category in GF-YR is represented in Table IV. We got the GF-5 HS and labeling data from the First Marine Institute, Department of Natural Resources, which was acquired over the Yellow River Delta National Nature Reserve in China.

In the experiment, the size of the overlapping area between HSI and MSI in Chikusei data is 1290 × 300, and in GF-YR

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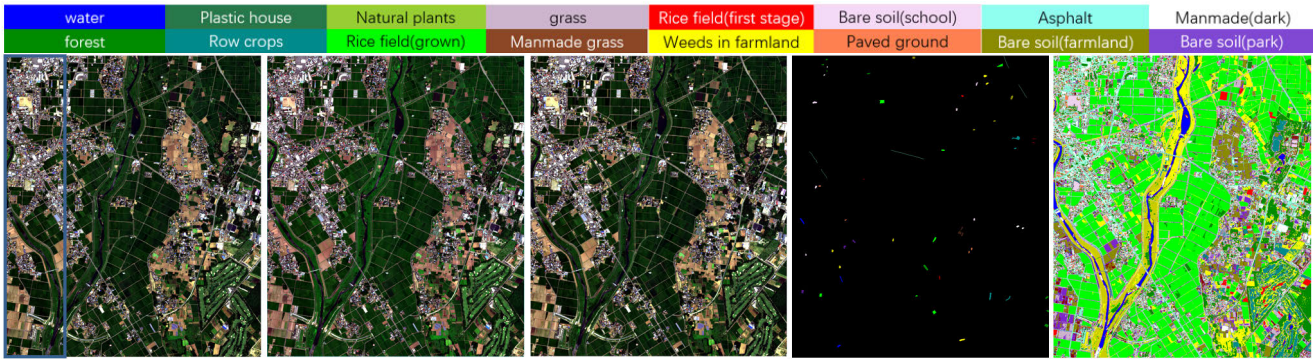


Fig. 2. Chikusei data. (a) Original HSI (displayed in 64, 42, 23 bands). (b) Chikusei HR-MSI data (displayed in 6, 4, 3 bands). (c) Chikusei LR-HSI data (displayed in 64, 42, 23 bands). Labels of (d) training set and (e) test set.

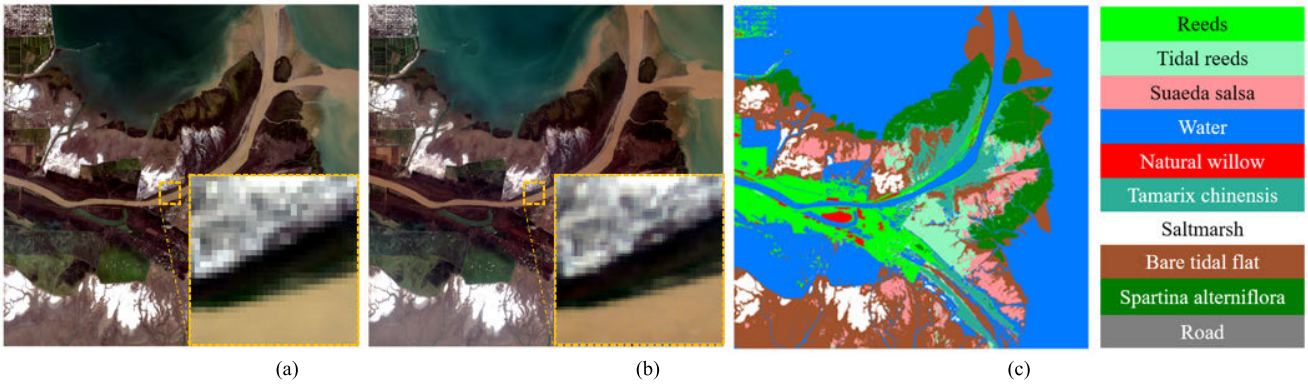


Fig. 3. GF-YR data. (a) LR-HSI acquired by GF-5 (displayed in 56, 39, 25 bands). (b) HR-MSI acquired by GF-1 (displayed in 3, 2, 1 bands). (c) Classification results for labeling data.

TABLE IV
NUMBER OF LABELS FOR GF-YR DATA

Labels	Categories	Number of global labels
1	Reeds	171779
2	Tamarix chinensis	104809
3	Tidal reeds	83161
4	Saltmarsh	76206
5	Suaeda salsa	102579
6	Naked tidal flat	436015
7	Water	1066570
8	Spartina alterniflora	180186
9	Natural willow	13492
10	Road	5371



Fig. 4. Overlapping data in the experiment.

data, the size is 500 × 500. As shown in Fig. 4, the overlapping area is indicated in the blue square.

B. Evaluation Indicators

We gauge the resemblance and the classification precision between the reconstructed HR-HSI and the original HSI in the experiment.

First, a range of supervised metrics are used to undergo a thorough evaluation of reconstruction similarity, which is applicable in scenarios where there is a referenceable ground truth. We use peak signal-to-noise ratio (PSNR) to measure the distortion after reconstruction, spectral angle index (SAM) and structural similarity (SSIM) are applied to measure spectral similarity and SSIM, respectively.

Second, unsupervised indicators such as spectral distortion index (D_λ), spatial distortion index (D_s), and global image quality (QNR) are introduced to evaluate the authenticity of the reconstructed images. Specifically, as there is no standard reference HR-HIS for the real dataset GF-YR, we can only apply unsupervised indicators to prove the authenticity and accuracy of the reconstructed image.

Third, we adopt overall accuracy (OA) and Kappa coefficient to assess the accuracy in classifying different features or classes.

The calculation formulas are as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{HWN}\sum_{l=1}^{N}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(u_l(i,j) - \widehat{u}_l(i,j)\right)^2}$$

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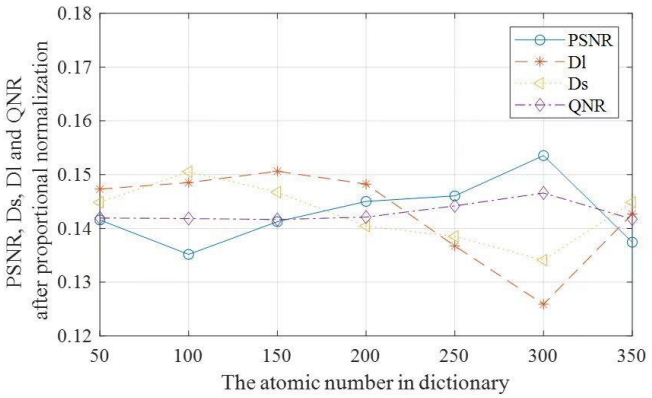


Fig. 5. Variation of evaluation metrics with the change of atomic numbers in the dictionary in Chikusei data.

$$\mathrm{PSNR} = 20 \lg \frac{1}{\mathrm{RMSE}} \tag{36}$$

where u represents the original HSI, $\hat{u}$ represents the reconstructed HSI. N, H, and W represent the number of bands, height, and width of HSI, respectively,

$$\mathrm{SAM} = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \arccos \left(\frac{\langle u(i,j), \hat{u}(i,j) \rangle}{|u(i,j)| \left|\hat{u}(i,j)\right|} \right) \tag{37}$$

$$\mathrm{SSIM} = \frac{(2\mu_u \mu_{\hat{u}} + c_1)(2\sigma_u \sigma_{\hat{u}} + c_2)}{(\mu_u^2 + \mu_{\hat{u}}^2 + c_1)(\sigma_u^2 + \sigma_{\hat{u}}^2 + c_2)}$$
$$c_1 = (k_1 L)^2, \quad c_2 = (k_2 L)^2, \quad c_3 = c_2/2. \tag{38}$$

c_1, c_2, c_3 are constants to avoid the denominator tending to a zero. Default $k_1 = 0.01, k_2 = 0.03$, L is the range of pixel values in the image

$$Q(\hat{u}_i, \hat{u}_j) = \frac{4\sigma_{\hat{u}_i \hat{u}_j} \bar{\hat{u}}_i \bar{\hat{u}}_j}{\left(\sigma_{\hat{u}_i}^2 + \sigma_{\hat{u}_j}^2\right)\left(\bar{\hat{u}}_i^2 + \bar{\hat{u}}_j^2\right)} \tag{39}$$

$$D_\lambda = \sqrt[q]{\frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{j=1, j\neq i}^{N} \left| Q(\hat{u}_i, \hat{u}_j) - Q(\mathbf{H}_i, \mathbf{H}_j) \right|^q} \tag{40}$$

$$D_s = \sqrt[q]{\frac{1}{N} \sum_{i=1}^{N} \left| Q(\hat{u}_i, \mathbf{M}) - Q\left(\mathbf{H}_i, \tilde{\mathbf{M}}\right) \right|^q} \tag{41}$$

$$\mathrm{QNR} = (1 - D_\lambda)^\mu (1 - D_s)^\rho \tag{42}$$

where $\mathbf{H}$ and $\mathbf{M}$ represent LR-HSI and HR-MSI, respectively. $\tilde{\mathbf{M}}$ is an MSI after the down-sampling of $\mathbf{M}$, which has the same spatial resolution as $\mathbf{H}$.

IV. Results and Analysis

As shown in Fig. 5, in Chikusei data, the influence of the atomic number of coupled dictionaries on the evaluation index of reconstructed HSI is displayed, and with the increase of the atomic number, the quality of the reconstructed image initially improves and then decreases. Subsequently, the optimal result is achieved when the dictionary atomic number is 300.

In the experiment, HR-HSI is reconstructed using the optimal dictionary atom number previously mentioned. Additionally, it is shown schematically for several bands of reconstructed HR-HSI and the comparison with original HSI is represented in Fig. 6. Across all selected bands, the spatial and spectral reconstruction quality is notably excellent. SVM classifier is used to classify and evaluate the reconstruction of Chikusei data. By randomly selecting 0.5% labeling data in GF-YR for training data, the KNN classifier is used to obtain a classification image of the whole data.

For comparison, specifically, we utilize J-SLoL [31], MPR-Net [42], MSSNet [43], PSJN [23], HSGAN [44], AWAN [45], and SSRNet [46]. Among these methods, J-SLoL belongs to the approach of dictionary learning, whereas the rest of the methods represent the method of deep learning. To ensure a fair comparison and highlight the effectiveness of the proposed method, in J-SLoL, we explored various atomic numbers within the dictionary and ultimately chose the image exhibiting the optimal reconstruction effect for index calculation. Additionally, in the training process of all of the deep learning methods, for Chikusei data, crop the overlapping area into several 64 × 64 blocks by randomly selecting a starting point, keeping a 50% repetition rate to ensure the coverage of the entire training area, and keep the number of blocks is 800.

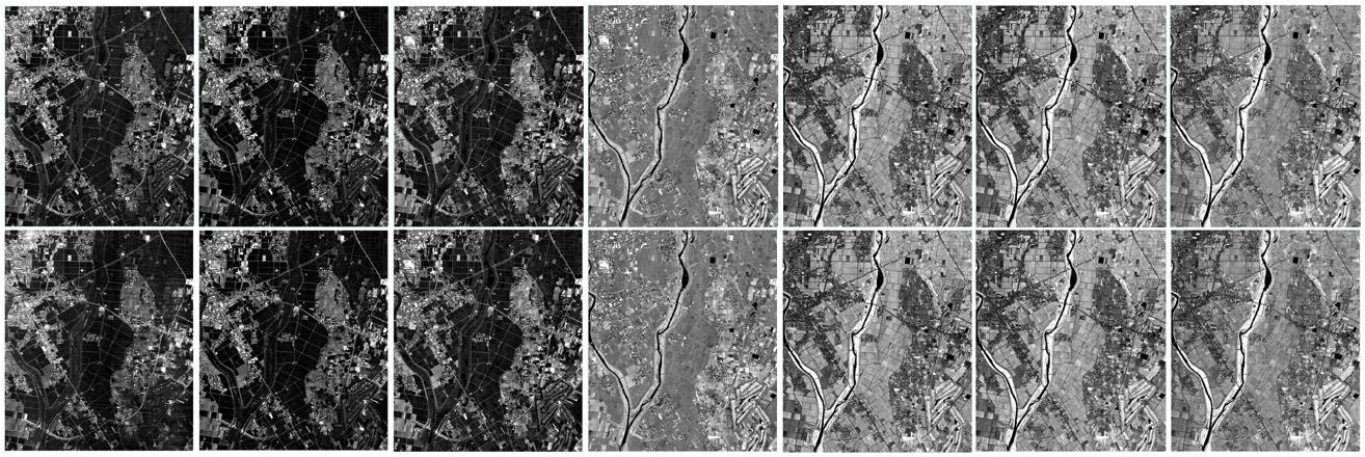
During the testing process, considering data volume and memory usage, the data is cropped to 32 image blocks with a size of 336 × 336. This segmentation results in a slight overlap among the cropped image blocks, ultimately yielding a composite image with dimensions of 1290 × 1508 pixels. The pixel values within the overlapping regions are ascertained by computing the average. Similarly, in GF-YR data, with the overlapping area of 500 × 500, the area is cropped into images of 256 × 256, maintaining a 50% repetition rate to ensure comprehensive coverage of the training area, we randomly select a starting point to designate the training field. Ultimately, the total of training blocks is 500. As for the testing process, the data is cropped into 256 × 256 with the number of 64, which leads to a total size of 1468 × 1526.

Tables V and VI compare evaluation metrics for reconstructed HR-HSI using various methods in Chikusei data and GF-YR data, respectively. The proposed Sub-CDL excels on Chikusei data, achieving optimal results in PSNR, OA, KAPPA, SSIM, and D_λ metrics. Specifically, PSNR has improved by 0.4223, OA has risen by 0.792%, and SSIM has improved by 0.0056. Notably, D_λ has reached the experimental minimum, and the QNR indicator is also close to the minimum value in the experiment, reflecting reduced spatial-spectral distortion and higher image quality.

Meanwhile, the unsupervised evaluation indexes in each method are consistent with supervised evaluation, indicating the effectiveness of unsupervised indexes. As for GF-YR data, proposed Sub-CDL has effective improvement on D_λ, QNR and OA. Specifically, D_λ has decreased by 0.0366, and QNR and OA have archived the maximum of the experimental methods.

To demonstrate the reconstructed HSIs clearly, as shown in Fig. 7, spectral curves of pixels at fixed spatial locations in various reconstruction HSIs in Chikusei data are plotted for

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(a) (b) (c) (d) (e) (f) (g)

Fig. 6. Recovery of each band (reconstructed HSI above and original HSI below), selecting (a) 4, (b) 27, (c) 46, (d) 68, (e) 79, (f) 94, and (g) 126 bands to display, respectively.

TABLE V

COMPARISON OF EVALUATION INDICATORS OF RECONSTRUCTION HSIS ON CHIKUSEI DATA

Methods	Sub-CDL	PSJN	MPRNet	HSGAN	AWAN	MSSNet	J-SLOL	SSRNet
PSNR	**38.5630**	38.1407	37.7841	37.9465	37.2593	36.0318	37.7805	37.3327
SSIM	**0.9752**	0.9488	0.9514	0.9404	0.9202	0.9472	0.9696	0.8742
SAM	1.8112	**1.5086**	3.8679	1.9076	2.2992	3.1497	1.8396	1.7988
D_λ	**0.0172**	0.0312	0.0545	0.0201	0.0455	0.0729	0.0370	0.0615
D_s	0.0387	**0.0249**	0.0404	0.0344	0.0292	0.1061	0.0449	0.0264
QNR	0.9448	0.9447	0.9073	**0.9461**	0.9267	0.8287	0.9198	0.9137
OA	**0.7826**	0.7764	0.7561	0.7598	0.7358	0.7481	0.7391	0.7462
KAPPA	**0.7420**	0.7351	0.7112	0.7189	0.6924	0.7038	0.6913	0.7039

TABLE VI

COMPARISON OF EVALUATION INDICATORS OF RECONSTRUCTION HSIS ON GF-YR DATA

Methods	Sub-CDL	PSJN	MPRNet	HSGAN	AWAN	MSSNet	J-SLOL	SSRNet
D_λ	**0.0872**	0.1599	0.1238	0.1598	0.1549	0.1614	0.1667	0.1597
D_s	0.1586	0.1950	0.1496	0.2223	0.2157	**0.1121**	0.1218	0.2131
QNR	**0.7679**	0.6763	0.7451	0.6534	0.6648	0.7444	0.7317	0.6612
OA	**0.7916**	0.7296	0.7726	0.7353	0.7023	0.7842	0.7815	0.7167
KAPPA	**0.7089**	0.6371	0.6887	0.6430	0.6027	0.7078	0.6960	0.6227

comparison. The blue curve represents the original HR-HSI, while different colors indicate different methods. Specifically, specific coordinates are chosen to demonstrate the results, including (50, 100), (600, 900), (500, 500), and (1400, 1200). Notably, compared to the contrast methods, the proposed Sub-CDL closely approximates the original HSI on Chikusei data. However, some of the methods exhibit offsets at certain coordinates, indicating imperfect spectral restoration in the selected range, possibly due to specific ground features. In the four regions, spectral curves of comparison methods exhibit shifts in several initial and final bands, whereas the proposed method consistently demonstrates excellent recovery in the range of 363–1018 nm.

Figs. 8 and 9 demonstrate the classification effect of each method in Chikusei and GF-YR data, respectively. Table VII compares reconstruction results in the test set of the Chikusei dataset for both single-class and overall classification accuracy. In Chikusei data, the original HR-HSI achieves 78.29% classification accuracy, while the original HR-MSI reaches 71.00%. The proposed Sub-CDL outperforms other reconstruction methods, achieving an overall classification accuracy of 78.26% on the test set, closely matching the HR-HSI, and archives a significant classification improvement compared to HR-MSI. The OA of the PSJN method scored 77.64%, and the vast majority of the other reconstruction methods are below 75%. Notably, Sub-CDL exhibits superior reconstruction results in classes Water, Paved ground, Bare soil, and Grass, and remains consistently stable classification accuracy across various other categories, indicating the effectiveness in preserving the image quality of the proposed method.

Similarly, in GF-YR data, as shown in Table VIII, Sub-CDL archives the highest classification accuracy among the three

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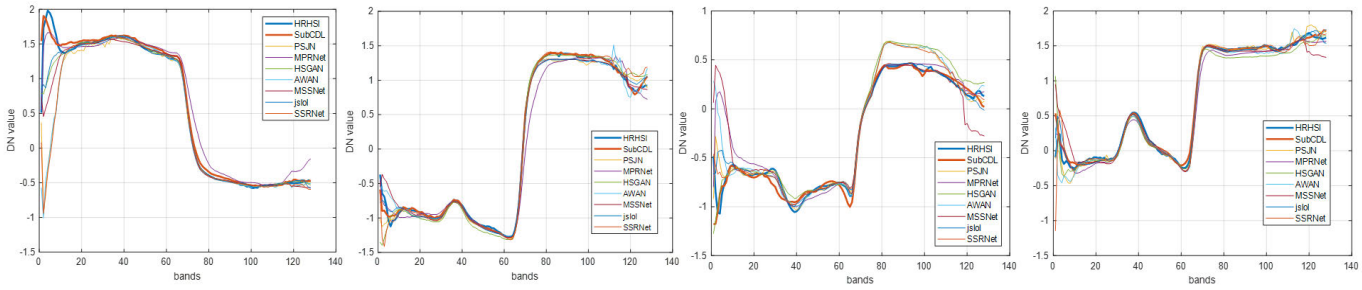


Fig. 7. Schematic illustration of the DN value of a fixed-position pixel in reconstructed HR-HSIs. (a) (50, 100). (b) (600, 900). (c) (500, 500). (d) (1400, 1200).

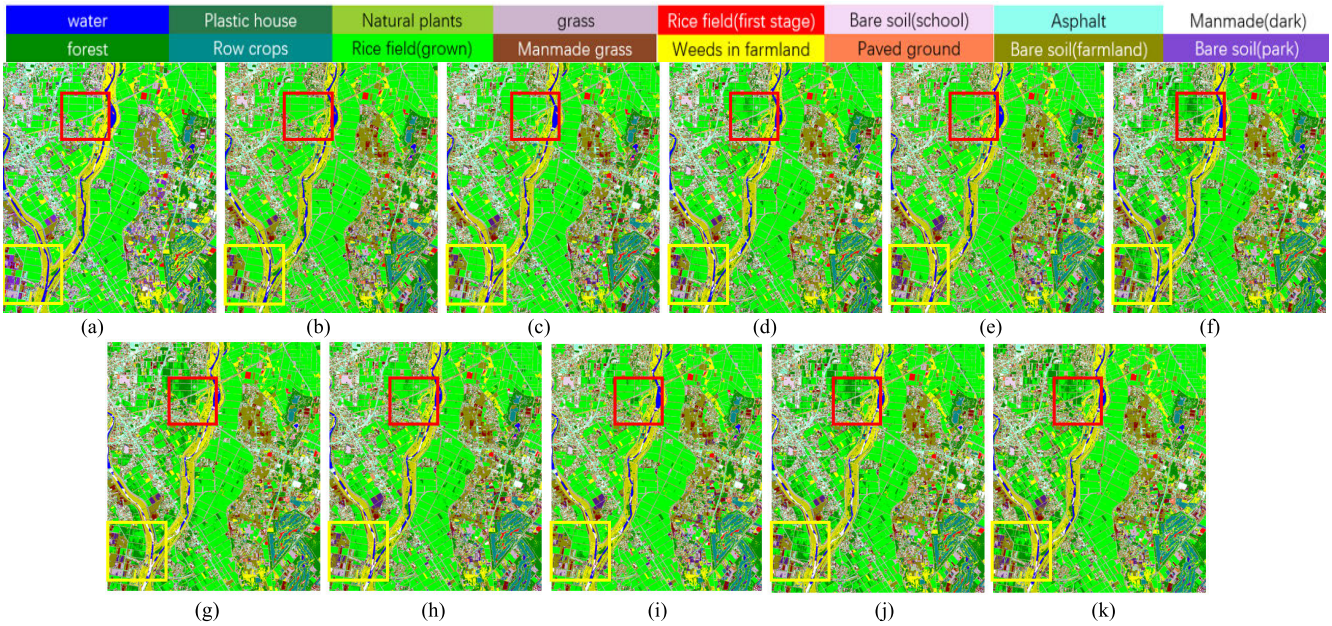


Fig. 8. Comparison of subpixel classification results of Chikusei data. (a) Ground truth. (b) Original HR-HSI. (c) Proposed Sub-CDL. (d) PSJN. (e) MPRNet. (f) HSGAN. (g) AWAN. (h) MSSNet. (i) J-SLOL. (j) SSRNet. (k) HR-MSI.

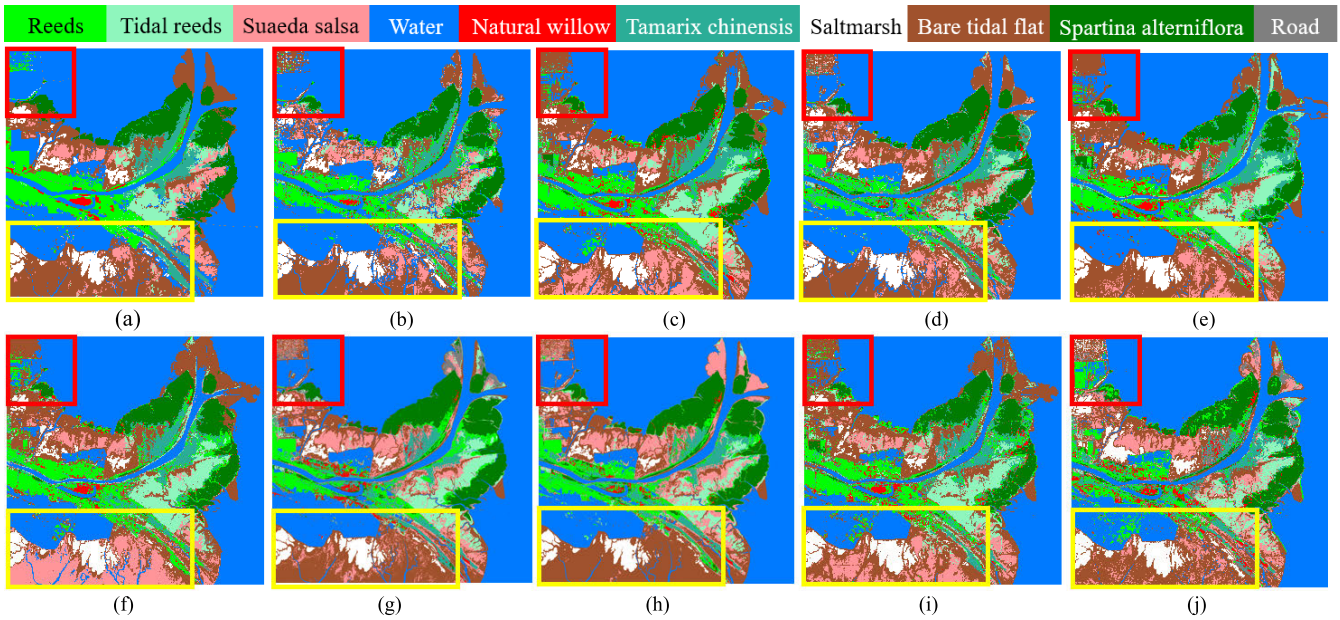


Fig. 9. Comparison of subpixel classification results of GF-YR data. (a) Labels. (b) Proposed Sub-CDL. (c) PSJN. (d) MPRNet. (e) HSGAN. (f) AWAN. (g) MSSNet. (h) J-SLOL. (i) SSRNet. (j) HR-MSI.

methods in classes Reeds, Suaeda salsa, and Water, which scored the highest OA to 79.16%, much higher than MSI's categorization. In the Tamarix chinensis and Bare tidal flat categories, the classification accuracy is highly comparable to,

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TABLE VII
COMPARISON OF CLASSIFICATION ACCURACY OF RECONSTRUCTION CHIKUSEI HR-HSIS BY DIFFERENT METHODS

	Categories	HRHSI	Sub-CDL	PSJN	MPRNet	HSGAN	AWAN	MSSNet	J-SLOL	SSRNet	HRMSI
1	Rice field (first stage)	0.8888	0.7205	0.8712	0.7182	**0.9226**	0.8258	0.7534	0.7354	0.8317	0.7505
2	Natural plant	0.7135	0.7182	0.6774	0.7150	**0.8016**	0.7508	0.7205	0.6766	0.7910	0.7444
3	Row crops	0.7557	0.8535	0.7455	0.6741	0.8707	**0.8718**	0.6930	0.7049	0.8504	0.7143
4	Asphalt	0.8191	0.8306	0.8104	0.8049	0.8372	0.8311	0.8136	0.8128	**0.8393**	0.8186
5	Forest	0.8232	0.7505	0.8155	**0.8796**	0.5164	0.4685	0.7072	0.7757	0.4822	0.4977
6	Water	0.9802	**1.0000**	0.9883	0.9875	0.9945	**1.0000**	0.9879	0.9850	**1.0000**	0.9836
7	Plastic house	**0.9665**	0.9125	0.9674	0.9563	0.9661	0.9564	0.9513	0.9444	0.9599	0.9449
8	Bare soil (school)	0.7198	0.7315	**0.7542**	0.6936	0.6990	0.7291	0.7085	0.7371	0.7332	0.7150
9	Bare soil (farmland)	0.3191	0.3578	0.3129	0.2651	0.3471	0.3380	0.2637	0.2469	**0.3594**	0.2736
10	Manmade grass	0.5137	0.5155	0.5006	0.4842	0.5250	0.5182	0.4938	0.4890	**0.5231**	0.5042
11	Paved ground	0.4096	**0.6019**	0.3898	0.4447	0.5051	0.4898	0.4564	0.4357	0.4896	0.4410
12	Manmade (dark)	**0.7843**	0.7083	0.7840	0.7172	0.7320	0.6866	0.6699	0.6912	0.6956	0.6936
13	Weeds in farm	**0.9317**	0.8614	0.9098	0.9121	0.8984	0.8550	0.9002	0.8914	0.8731	0.8798
14	Bare soil (park)	0.8480	**0.8531**	**0.8531**	0.6485	0.8039	0.8299	0.5937	0.5627	0.8403	0.6815
15	grass	0.8016	**0.8384**	0.8023	0.7470	0.8355	0.8032	0.7606	0.7553	0.8082	0.7645
16	Rice field (grown)	0.9437	0.9332	0.9456	0.9412	0.9490	0.9522	**0.9608**	0.9291	0.9468	0.9090
	OA	**0.7829**	0.7826	0.7764	0.7561	0.7598	0.7358	0.7481	0.7391	0.7462	0.7100
	KAPPA	**0.7426**	0.7420	0.7351	0.7112	0.7189	0.6924	0.7038	0.6913	0.7039	0.6607

TABLE VIII
COMPARISON OF CLASSIFICATION ACCURACY OF RECONSTRUCTION GF-YR HR-HSIS BY DIFFERENT METHODS

	Categories	MSI	Sub-CDL	PSJN	MPRNet	HSGAN	AWAN	MSSNet	J-SLoL	SSRNet
1	Reeds	0.3606	**0.5850**	0.5557	0.5030	0.5387	0.5841	0.5786	0.5221	0.5568
2	Tamarix chinensis	0.4891	0.5965	0.5979	0.4588	0.5332	0.4874	0.4674	0.4631	**0.6045**
3	Tidal reeds	0.2802	0.4996	**0.6762**	0.4369	0.6221	0.6558	0.4478	0.4300	0.6683
4	Saltmarsh	**0.9416**	0.8402	0.7995	0.8217	0.7632	0.8060	0.8311	0.8372	0.7825
5	Suaeda salsa	0.4214	**0.4999**	0.4383	0.4701	0.3615	0.4367	0.4357	0.4059	0.4253
6	Bare tidal flat	0.6147	0.6801	0.6390	**0.7306**	0.7230	0.5166	0.6755	0.6466	0.6430
7	Water	0.8633	**0.9577**	0.8421	0.8745	0.8517	0.8328	0.9510	0.9565	0.8175
8	Spartina alterniflora	0.7344	0.8684	0.7305	0.8035	0.6824	0.7848	**0.8876**	0.8818	0.7211
9	Nature willow	0.4252	0.3001	0.4530	0.1235	**0.4762**	0.4623	0.4229	0.3693	0.4327
10	Road	0.0413	0.0350	**0.1661**	0.0150	0.1127	0.1633	0.0451	0.0309	0.1588
	OA	0.6970	**0.7916**	0.7296	0.7726	0.7353	0.7023	0.7842	0.7815	0.7167
	KAPPA	0.5860	**0.7089**	0.6371	0.6887	0.6430	0.6027	0.7078	0.6960	0.6227

or nearly approximates, the optimal values attained by these methodologies. Accordingly, the proposed Sub-CDL has a universal categorization advantage across diverse feature types in both Chikusei and GF-YR data.

In proposed Sub-CDL, the overall running time amounts to approximately 1872 s. In comparison, the running time of another dictionary learning-based program J-SLOL is approximately 1396 s, slightly faster than the proposed Sub-CDL. However, in terms of the authenticity and accuracy of reconstructed HR-HSI, Sub-CDL is significantly outperforms J-SLOL. Compared with other deep learning methods, these deep learning-based methods generally exhibit shorter testing durations, typically within 180 s. Conversely, the training time they require is very long, and it takes several days to obtain the results as shown in the tables of the manuscript.

V. CONCLUSION

In this article, we innovatively propose a sub-pixel coupled dictionary method called Sub-CDL, leveraging sub-pixel representation for reconstructing large-scene and high-spatial-resolution HR-HSI using overlapping regions of LR-HSI and HR-MSI. Minimizing spatial and spectral reconstruction distortions, Sub-CDL enhanced the image authenticity and classification accuracy of reconstructed HR-HSIs. Innovatively, by incorporating the sparse representation coefficient derived from HR-MSI, a correlation between different spatial resolutions is established by integrating the spatial down-sampling matrix and the spatial blurred matrix into the model, a closer fit for real HSIs can be achieved and the accuracy of spatial recovery is further enhanced. SRF is applied to resolve spectral disparities in input images. Furthermore, spatial regularization constraints and sub-pixel coupled dictionary low-rank constraints are utilized to retain edges and detailed spatial information in images and enhance the precision of learned coupled dictionaries. Experimental results demonstrate that the reconstructed HR-HSIs exhibit superior image authenticity and classification accuracy at a range of 363–2500 nm, while the distortions of reconstructed images are minimized.

Yet, the method requires necessitates manual parameter selection, potentially complicating the computational process. Consequently, the adoption of a multi-objective optimization approach can be considered to eliminate the need for manual selection of intricate parameters.

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